

# END-TO-END IRIS FLOWER CLASSIFICATION

## Using Multinomial Logistic Regression

A Machine Learning Project Report

### 1. Problem Statement & Motivation

#### The Problem

In botanical research and automated agriculture, manually identifying plant species based on physical dimensions is time-consuming and prone to human error. Machine learning provides a fast, reproducible way to classify plant varieties automatically using measured morphological attributes.

#### Why Solve This?

The Iris Flower Dataset serves as a benchmark problem in pattern recognition. The core challenge is to accurately differentiate between three related species (*Iris-setosa*, *Iris-versicolor*, and *Iris-virginica*) using four simple measurements:

- Sepal Length (cm)
- Sepal Width (cm)
- Petal Length (cm)
- Petal Width (cm)

While *Iris-setosa* is easily distinguishable, *Iris-versicolor* and *Iris-virginica* share overlapping physical characteristics, creating a realistic classification challenge. Solving this using a statistical linear model (Multinomial Logistic Regression) demonstrates how far clean data preparation can push simpler, highly interpretable algorithms before resorting to complex, “black-box” models.

### 2. Comprehensive Pipeline & Methodology

Rather than feeding raw data directly into a model, a structured data engineering workflow was implemented to guarantee data integrity, avoid leakage, and maximize model generalization:

|   |                                                                                                            |
|---|------------------------------------------------------------------------------------------------------------|
| 1 | <b>Data Auditing &amp; Cleaning</b><br>Checked for null values and removed identical duplicate rows.       |
| 2 | <b>IQR Outlier Management</b><br>Capped extreme values at percentile fences using <code>np.clip()</code> . |

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**Stratified Train/Test Split**

80/20 split preserving the 1:1:1 class distribution.

4

**Feature Standardization**

Scaled features to zero mean and unit variance.

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**Softmax Regression Fitting**

Trained Multinomial Logistic Regression (L-BFGS solver).

6

**Evaluation & Discussion**

Analyzed precision, recall, F1-score, and the confusion matrix.

### Step 1: Structural Data Auditing & Duplicate Removal

**Missing Value Check:** Audited all four continuous features. Zero null or missing values were found, ensuring no imputation was needed.

**Duplicate Removal:** Identified identical row entries resulting from measurement rounding. Dropping these duplicates (150 → 147 total records) was crucial to prevent identical samples from sitting in both the training and test sets simultaneously (data leakage).

### Step 2: Outlier Detection & IQR Percentile Capping

Instead of deleting records with extreme values — which reduces dataset size — an Interquartile Range (IQR) capping strategy was executed:

- First-quartile ( $Q_1$ ) and third-quartile ( $Q_3$ ) bounds were calculated for each continuous feature.
- Upper and lower fences were established as: Lower Fence =  $Q_1 - 1.5 \times \text{IQR}$ , Upper Fence =  $Q_3 + 1.5 \times \text{IQR}$ .
- Any feature values exceeding these fences (primarily observed in SepalWidthCm, with 1 low-side and 3 high-side outliers) were capped at the boundary limits using `np.clip()`.
- Validation: Post-capping verification via `min()/max()` comparison confirmed all extreme points were pulled safely back inside the mathematical bounds without dropping a single sample.

### Step 3: Stratified Train-Test Splitting

To prevent class imbalance across splits, a Stratified 80/20 Train-Test Split was applied ( $N_{\text{Train}} = 117$ ,  $N_{\text{Test}} = 30$ ). Stratification ensures that the exact 1:1:1 ratio of Setosa, Versicolor, and Virginica samples exists in both the training set and the unseen testing set.

### Step 4: Feature Standardization

Because Logistic Regression relies on gradient-based optimization solvers (lbfgs), features with larger natural scales can dominate weight updates. `StandardScaler` was fitted strictly on the training data and then applied to both sets, using the transformation  $z = (x - \mu) / \sigma$ . This converted all physical dimensions to a shared standard scale with a mean of 0 and a standard deviation of 1.

Step 5: Model Optimization (Softmax / Multinomial Logistic Regression)

Because this is a 3-class target, standard binary logistic regression was extended to Multinomial Logistic Regression. The model calculates class probabilities simultaneously using the Softmax function, optimizing multi-class log-loss across all three categories at once:

$$P(Y = k | x) = \frac{\exp(w_k^T x + b_k)}{\sum_j \exp(w_j^T x + b_j)}$$

3. Detailed Results & Performance Analysis

The final model was evaluated strictly on the unseen 20% test dataset (30 samples total).

Overall Test Accuracy: 93.33%

Comprehensive Performance Table

| Species Class    | Precision | Recall | F1-Score | Support |
|------------------|-----------|--------|----------|---------|
| Iris-setosa      | 1.00      | 1.00   | 1.00     | 10      |
| Iris-versicolor  | 0.90      | 0.90   | 0.90     | 10      |
| Iris-virginica   | 0.90      | 0.90   | 0.90     | 10      |
| Macro Average    | 0.93      | 0.93   | 0.93     | 30      |
| Weighted Average | 0.93      | 0.93   | 0.93     | 30      |

4. In-Depth Discussion & Findings

1. Perfect Classification on Iris-setosa (1.00 F1-Score)

**Finding:** The model achieved 100% precision and recall on Iris-setosa (10/10 test samples correct).

**Domain Explanation:** This is not overfitting. In feature space, Setosa petal dimensions sit in an isolated cluster far away from the other two species. It is linearly separable, meaning a straight decision plane easily isolates Setosa with zero classification error.

2. Boundary Overlap Between Versicolor and Virginica (0.90 F1-Score)

**Finding:** The model misclassified exactly 1 Iris-versicolor sample as virginica, and 1 Iris-virginica sample as versicolor.

**Domain Explanation:** Physical attributes of versicolor and virginica naturally blend together at the higher end of petal dimensions. Because Logistic Regression constructs straight decision boundaries, these few boundary samples fall on the wrong side of the linear line.

3. Takeaway on Model Selection

Achieving 93.33% test accuracy proves that lightweight, interpretable linear models are exceptionally effective when paired with proper data cleaning, duplicate removal, and outlier management — eliminating the need for complex ensemble models for this classification task.

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*Prepared as part of an academic machine learning project.*